

ERTIS, Kazakhstan

WMO: 298070

Lat: 53.3375N

Lon: 75.4619E

Elev: 94

StdP: 100.20

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-34.6	-31.5	-37.6	0.1	-34.5	-34.7	0.1	-31.6	12.1	-10.0	10.0	-11.0	1.3	30	0.571

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	32.0	18.3	30.1	17.9	28.4	17.4	20.0	27.9	19.1	27.1	18.4	26.2	2.7	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.3	12.5	22.8	16.2	11.7	22.4	15.2	10.9	22.0	57.7	27.9	55.0	27.0	52.4	26.0	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.0	9.1	7.7	DB	-36.9	35.1	3.8	2.0	-39.6	36.5	-41.8	37.7	-43.9	38.9	-46.6	40.4	(4)
(5)			WB	-37.0	22.1	3.7	0.9	-39.6	22.8	-41.7	23.3	-43.8	23.8	-46.4	24.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.0	-18.0	-15.5	-7.0	5.9	13.6	19.3	20.6	18.4	12.2	4.5	-5.8	-14.0	(6)	
	DBStd	15.00	8.00	7.21	6.65	5.94	5.27	4.22	3.59	3.98	4.72	5.05	7.73	7.74	(7)	
	HDD10.0	3712	867	713	528	149	26	1	0	1	29	181	474	743	(8)	
	HDD18.3	5847	1126	946	786	373	163	39	18	49	193	428	724	1001	(9)	
	CDD10.0	1139	0	0	0	26	137	281	329	262	94	11	0	0	(10)	
	CDD18.3	234	0	0	0	1	16	68	90	52	7	0	0	0	(11)	
	CDH23.3	2413	0	0	0	12	230	687	844	536	102	2	0	0	(12)	
	CDH26.7	792	0	0	0	2	61	228	292	180	29	0	0	0	(13)	
(14) Wind	WSAvg	2.8	2.7	3.0	3.1	3.2	3.2	2.5	2.1	2.2	2.5	2.6	3.0	3.1	(14)	
(15) Precipitation	PrecAvg	293	14	11	11	14	26	39	55	40	21	26	20	16	(15)	
	PrecMax	465	41	29	34	38	74	98	206	78	65	61	62	35	(16)	
	PrecMin	147	1	0	1	0	0	4	8	9	3	7	1	1	(17)	
	PrecStd	70	8	7	8	10	17	24	42	20	13	14	11	8	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.3	1.4	9.5	25.2	31.2	33.4	34.5	33.9	29.9	21.5	9.7	1.5	(19)	
		MCWB	0.1	0.0	5.9	14.9	17.1	18.7	19.0	18.4	16.8	12.3	6.7	0.2	(20)	
	2%	DB	-1.4	-0.2	5.4	21.4	28.4	31.3	31.8	30.9	26.1	18.0	6.9	0.4	(21)	
		MCWB	-2.4	-1.4	3.0	12.9	16.0	18.2	18.3	18.1	15.5	10.8	4.7	-0.7	(22)	
	5%	DB	-3.7	-2.5	3.1	18.6	26.0	29.3	29.9	28.5	23.2	14.9	4.8	-1.2	(23)	
		MCWB	-4.8	-3.7	1.2	11.5	15.2	17.7	18.2	17.6	14.2	9.1	2.7	-2.3	(24)	
	10%	DB	-5.9	-4.8	1.6	15.6	23.4	27.4	27.9	26.4	20.2	12.1	2.8	-3.2	(25)	
		MCWB	-6.8	-5.9	0.0	9.9	14.1	16.9	17.9	17.0	12.9	7.7	1.1	-4.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.0	0.1	6.0	15.5	18.4	20.9	21.3	20.4	18.0	12.8	7.1	0.5	(27)	
		MCDB	1.1	1.3	9.0	23.5	28.6	30.3	28.9	28.9	27.1	20.2	9.0	1.2	(28)	
	2%	WB	-2.4	-1.5	3.1	13.4	16.8	19.5	20.2	19.3	16.1	11.0	4.8	-0.6	(29)	
		MCDB	-1.4	-0.3	5.2	20.4	26.4	27.7	27.7	28.2	24.9	17.2	6.9	0.3	(30)	
	5%	WB	-4.8	-3.7	1.3	11.8	15.7	18.6	19.4	18.4	14.6	9.4	2.8	-2.4	(31)	
		MCDB	-3.8	-2.6	2.9	18.2	24.6	26.7	26.8	26.7	22.2	14.3	4.6	-1.3	(32)	
	10%	WB	-6.8	-5.9	0.1	10.2	14.5	17.7	18.6	17.4	13.3	7.9	1.1	-4.3	(33)	
		MCDB	-5.9	-4.8	1.6	15.3	22.6	25.7	25.8	24.9	19.7	11.9	2.7	-3.2	(34)	
(35) Mean Daily Temperature Range	MDBR		8.1	8.7	8.8	10.9	12.4	11.3	10.8	11.3	11.1	9.0	7.1	7.9	(35)	
	5% DB	MCDBR	8.3	7.9	7.8	15.5	15.9	14.0	13.7	14.6	15.3	14.3	7.9	8.0	(36)	
		MCWBR	8.0	7.4	6.4	8.9	7.4	5.5	5.0	6.0	7.5	8.7	6.3	7.6	(37)	
	5% WB	MCDBR	8.2	8.1	7.9	15.0	14.5	12.2	11.6	13.3	14.5	13.0	7.5	8.1	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.295	0.326	0.385	0.410	0.411	0.419	0.441	0.428	0.373	0.353	0.333	0.300	(40)	
	taud		2.040	1.993	1.934	2.140	2.198	2.257	2.198	2.205	2.309	2.257	2.118	2.014	(41)	
	Ebn at Noon		633	737	769	815	837	833	804	790	793	716	583	535	(42)	
	Edh at Noon		75	110	145	135	134	128	134	126	100	85	71	64	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.13	2.19	3.64	4.64	5.87	6.22	5.64	4.79	3.49	1.86	0.89	0.73	(44)	
	RadStd		0.09	0.09	0.31	0.39	0.50	0.33	0.42	0.34	0.30	0.24	0.11	0.10	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	+0.60	+0.84	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page